

Diya Kalra

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EDUCATION

Thapar Institute of Engineering and Technology

Bachelor of Engineering in Computer Science, CGPA: 9.59

Patiala, Punjab

2023 – 2027

OSDAV Public School

Senior Secondary (CBSE), Non-Medical – 96.6%

Kaithal, Haryana

2023

OSDAV Public School

Secondary (CBSE) – 97.6%

Kaithal, Haryana

2021

ACHIEVEMENTS

Meta PyTorch OpenEnv Hackathon 2026

- Selected among the **Top 2.6% teams** from **31,000+ registrations** nationwide; built **AMASES**, a multi-agent reinforcement learning environment using the OpenEnv framework.

Advanced Programming Camp by Codeforces Master

- Selected among **60,000+** students nationwide as a mentee for an Advanced Programming Camp led by a Codeforces Master; learned advanced algorithms including Matrix Exponentiation and Square Root Decomposition, solved two advanced challenge problems, and earned Hall of Fame recognition.

Merit Scholarships – Thapar Institute of Engineering & Technology

- Awarded **Merit-I and Merit-II Scholarships** worth **6 lakh rupees** for outstanding academic performance across multiple semesters.

PROJECTS

MedAssist

Evidence-Aware Medical Information Copilot — Flask, LangChain, Gemini, Pinecone, CrossEncoder, RAG

- Built an RAG pipeline to improve retrieval quality and reduce unnecessary LLM calls for medical queries.
- Integrated semantic routing, Pinecone retrieval, CrossEncoder reranking, and Gemini for evidence-grounded response generation.
- Reduced unnecessary **LLM invocations by 52%**, cutting end-to-end latency from **1050 ms** to **508 ms**, and achieved **78% routing accuracy** on a **100-query benchmark**.



Local Language Integrator

Real-Time Multilingual Chat Platform — React, FastAPI, Socket.IO, Firebase Firestore

- Enabled real-time communication supporting **14+ Indian languages** with automatic message translation.
- Implemented Socket.IO-based messaging with typing indicators, online presence, persistent chat history using Firebase Firestore, and deployed the application on Vercel and Render.



AMASES

Adaptive Multi-Agent Skill Evolution System — Python, OpenEnv, TRL-GRPO, Hugging Face, Reinforcement Learning

- Built an adaptive multi-agent reinforcement learning environment enabling curriculum-driven collaborative, competitive, teaching, and debate-based skill evolution.
- Implemented a TRL-GRPO training pipeline with Qwen and Llama models, adaptive curriculum learning, and structured reward engineering across **15 benchmark tasks (5 task families × 3 difficulty levels)**.
- Designed reward tracking and skill evolution analytics to evaluate policy learning and stable multi-agent training.



TECHNICAL SKILLS

Programming Languages: Python, C++, C, SQL

Machine Learning: Scikit-learn, NumPy, Pandas, Matplotlib

Generative AI: LangChain, Retrieval-Augmented Generation (RAG), Hugging Face Transformers, AI Agents, Prompt Engineering

Databases & Vector Stores: Pinecone, Firebase Firestore

Web Development: Flask, Next.js, React.js, Firebase, Socket.IO, REST APIs

Core CS Subjects: Data Structures Algorithms, Object-Oriented Programming (OOP), Operating Systems, Database Management Systems (DBMS), Computer Networks